Case Study: Reducing Late Deliveries Across the Dark Store Network

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1. The problem

Across the network, **62.6% of orders missed their promised delivery window**, not an isolated issue, but a systemic one. This matters because delivery speed is the core value proposition in quick commerce; sustained lateness at this scale puts customer trust and retention at risk before it shows up as a churn number.

The question wasn't *"are we late?"* It clearly was. The real question was: **where, when, and why** is this happening, so the fix can be targeted instead of generic.

2. What the data showed

Working from cleaned order-level data (duplicates removed, missing values handled, delivery times validated), three patterns stood out:

A. The problem is concentrated in peak hours, not spread evenly. Late rate sits around 45-53% during off-peak hours, but **spikes to 90-96% during lunch (12-1 PM) and dinner (7-10 PM)**. This isn't random variation; it's demand surge outpacing fulfillment capacity at predictable times.

B. Store performance varies meaningfully, even within the same city. Mumbai's three dark stores range from 61% to 70% late, a 9-point spread within one city. Since demand patterns are similar across stores in the same city, this points to **store-level execution gaps** (staffing, layout, picking process) rather than purely a demand problem.

C. Customer experience holds up until a threshold, then breaks. Average rating barely moves for orders 0-10 minutes late (4.26-4.27). But orders **10+ minutes late drop average rating to 4.16**, a real, identifiable threshold worth protecting against, not a gradual decline.

Specific hotspots (e.g., several store-hour combinations running at or near 100% late) are documented in the dashboard and SQL results, giving operations teams an exact list of where to act first.

3. Recommendation

Don't fix this with a blanket policy. Fix it where the data says it's breaking:

1. **Dynamic peak-hour staffing/dispatch buffer** for the identified 12-1 PM and 7-10 PM windows, store-by-store, rather than a uniform network-wide rule. This directly targets the 90%+ late spike without overcorrecting at off-peak hours.

2. **Store-level operational review for consistent underperformers** (e.g., MUM-DS-01, MUM-DS-02), since their gap versus peer stores in the same city points to a local, fixable process issue rather than demand.
3. **Automated late-delivery alerting**, a lightweight rule that flags any order trending past the 10-minute-late threshold in real time, since that's the point where customer experience measurably drops. This turns a lagging metric (rating, reviewed weekly) into a leading one (delay, caught mid-delivery).

4. Why this approach

This reflects how I'd want to work on the Operation Support / Technical team: **start from the data, not assumptions**, find where the problem actually concentrates, separate demand-driven issues from execution-driven ones, and recommend fixes precise enough to act on immediately rather than broad statements that sound right but don't move the metric.

Full dataset, cleaning script, SQL queries, and interactive dashboard available in the project repository.